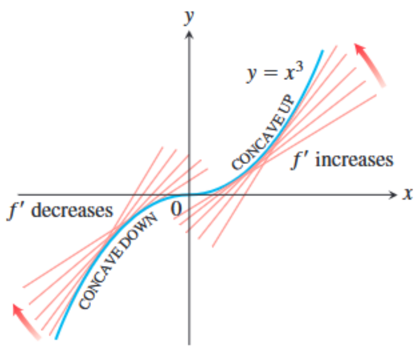
4.4 Concavity and Curve Sketching

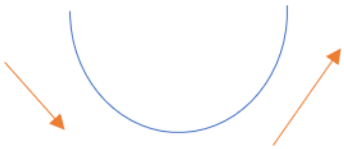
We have seen how the first derivative tells us where a function is increasing, where it is decreasing, and whether a local maximum or local minimum occurs at a critical point. In this section we see that the second derivative gives us information about how the graph of a differentiable function bends or turns. With this knowledge about the first and second derivatives, coupled with our previous understanding of symmetry and asymptotic behavior studied in Sections 1.1 and 2.6, we can now draw an accurate graph of a function. By organizing all of these ideas into a coherent procedure, we give a method for sketching graphs and revealing visually the key features of functions. Identifying and knowing the locations of these features is of major importance in mathematics and its applications to science and engineering, especially in the graphical analysis and interpretation of data. When the domain of a function is not a finite closed interval, sketching a graph helps to determine whether absolute maxima or absolute minima exist and, if they do exist, where they are located.

① Concavity

As you can see in the following Figure,



the curve $y = x^3$ rises as x increases, but the portions defined on the intervals $(-\infty, 0)$ and $(0, \infty)$ turn in different ways. As we approach the origin from the left along the curve, the curve turns to our right and falls below its tangents. The slopes of the tangents are decreasing on the interval $(-\infty, 0)$. As we move away from the origin along the curve to the right, the curve turns to our left and rises above its tangents. The slopes of the tangents are increasing on the interval $(0, \infty)$. This turning or bending behavior defines the concavity of the curve. The followings are the two different concavities.



Concave up

Concave down

DEFINITION The graph of a differentiable function $y = f(x)$ is

(a) concave up on an open interval I if f' is increasing on I;

(b) concave down on an open interval I if f' is decreasing on I.

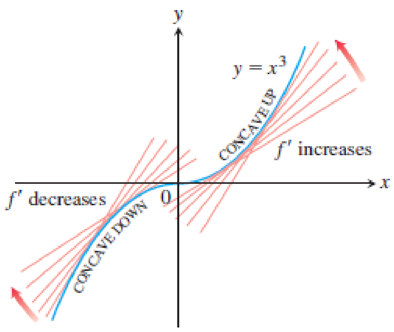
② The Second Derivative Test for Concavity

◆ Let $y = f(x)$ be twice-differentiable on an interval I.

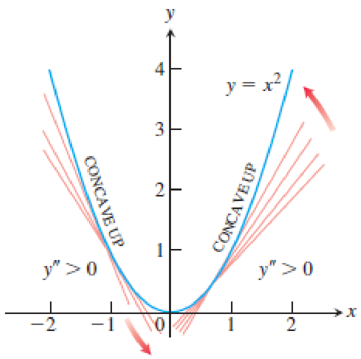
1. If $f'' > 0$ on I, the graph of f over I is concave up.
2. If $f'' < 0$ on I, the graph of f over I is concave down.

Example 1

(a) The curve $y = x^3$ is concave down on $(-\infty, 0)$, where $y'' = 6x < 0$, and concave up on $(0, \infty)$, where $y'' = 6x > 0$.

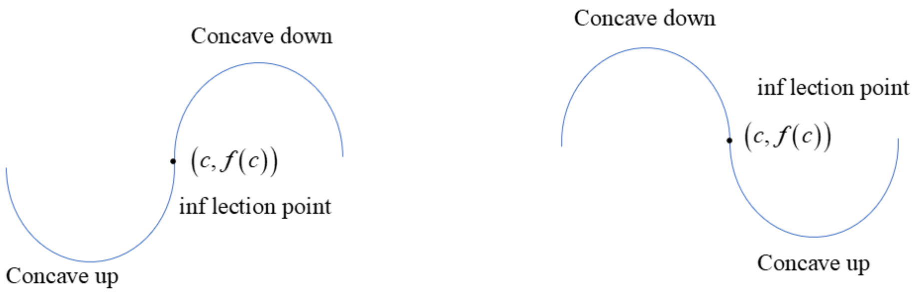


(b) The curve $y = x^2$ is concave up on $(-\infty,\ \infty)$ because its second derivative $y'' = 2$ is always positive.



Example 2 Determine the concavity of $y = 3 + \sin x$ on $[0, 2\pi]$.

③ Points of Inflection



DEFINITION A point $(c,\ f(c))$ where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**.

Note 1 At a point of inflection $(c,\ f(c))$, either $f''(c) = 0$ or $f''(c)$ fails to exist.

Note 2 $f''(c)$ exists and $f(x)$ has a inflection point at $x = c \Leftrightarrow f''(c) = 0$ and $f''(x)$ has different sign at two sides of c.

Note 3 $f''(c)$ doesn't exists and $f(x)$ has an inflection point at $x = c \Leftrightarrow f''(x)$ has different sign at two sides of c.

Example 3 Determine the concavity and find the inflection points of the function

$$f(x) = x^3 - 3x^2 + 2.$$

Example 4 Find the inflection point of $f(x) = x^{\frac{5}{3}}$.

Example 5 The curve $y = x^4$ has no inflection point at $x = 0$.

In the next example a point of inflection occurs at a vertical tangent to the curve where neither the first nor the second derivative exists.

Example 6 The graph of $y = x^{\frac{1}{3}}$ has a point of inflection at the origin because the second derivative is positive for $x < 0$ and negative for $x > 0$:

$$y'' = \frac{d^2}{dx^2}\left(x^{\frac{1}{3}}\right) = \frac{d}{dx}\left(\frac{1}{3}x^{-\frac{2}{3}}\right) = -\frac{2}{9}x^{-\frac{5}{3}},\ x \neq 0.$$

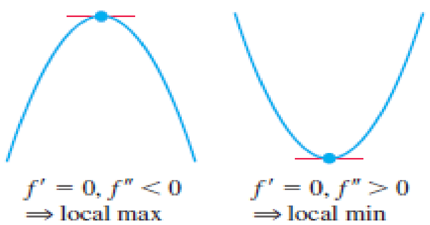
However, both y' and y'' fail to exist at $x = 0$, and there is a vertical tangent there.

④ Second Derivative Test for Local Extrema

THEOREM 5 Second Derivative Test for Local Extrema

Suppose f'' is continuous on an open interval that contains $x = c$.

1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.



◆ Sketch a graph of the function that captures its key features:

Suppose that $f(x)$ is continuous on its domain, and f' and f'' exist. To sketch the graph of $f(x)$

Step 1: Find f' and f''.

Step 2: Find the critical points of f, if any, and determine where the curve is increasing and where it is decreasing.

Step 3: Find the points of inflection, if any occur, and determine the concavity of the curve.

Step 4: Identify any asymptotes that may exist.

Step 5: Plot key points, such as the intercepts and the points found in Steps 2–3, and sketch the curve together with any asymptotes that exist.

Example 7 Sketch a graph of the function

$$f(x) = x^4 - 4x^3 + 10$$

using the above steps.

Example 8 Sketch the graph of $f(x) = \frac{(x+1)^2}{1+x^2}$.

Example 9 Sketch the graph of $f(x) = \frac{x^2+4}{2x}$ $(D = \{x \neq 0\})$.